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Preface

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List of Notations

The following table lists the notation used throughout this report, organized by topic.

Symbol	**Meaning**
General	
$\mathbb{N}$	Set of all natural numbers i.e., positive integers
$\mathbb{Z}$	Set of all integers
$\mathbb{Z}_0$	Set of all non-negative integers
$\mathbb{R}$	Set of real numbers
$\mathbb{C}$	Set of complex numbers
$[n]$	The set $\{1, 2, \ldots, n\}$ for $n \in \mathbb{N}$
i	Imaginary unit: $i^2 = -1$
S^1	$\{z \in \mathbb{C} : \lvert z\rvert = 1\}$, the multiplicative group of unit complex numbers
ζ_m	A primitive m-th root of unity
$[x]$	Floor function: largest integer $\leq x$; also written $\lfloor x \rfloor$
Finite Fields	
$\mathbb{F}_q$	Finite field with q elements
$\mathbb{F}_q^*$	Multiplicative group of nonzero elements of $\mathbb{F}_q$
$\mathbb{F}_p$	Prime field with p elements, $p = \mathrm{char}(\mathbb{F}_q)$
$\mathbb{F}_q^{*\,(d)}$	Subgroup of d-th powers in $\mathbb{F}_q^*$
E, F, L, K	Generic notation for finite fields or finite extensions of $\mathbb{F}_q$
$[E : \mathbb{F}_q]$	Degree of the extension $E/\mathbb{F}_q$
$\mathbb{F}_q[X_1, \ldots, X_n]$	Polynomial ring in n variables over $\mathbb{F}_q$
$\mathbb{F}_q^{\leq d}[X_1, \ldots, X_n]$	Polynomials in $\mathbb{F}_q[X_1, \ldots, X_n]$ of total degree at most d
Characters of Finite Abelian Groups	
G	A finite Abelian group with identity element 1_G
Z_k	Cyclic multiplicative group of order k
χ_0	Trivial character of G: $\chi_0(g) = 1$ for all $g \in G$
$\overline{\chi}$	Conjugate (and inverse) character of χ: $\overline{\chi}(g) = \overline{\chi(g)}$
$\mathrm{Hom}(A, B)$	Set of all group homomorphisms from A to B
$\widehat{G} = \mathrm{Hom}(G, S^1)$	Dual group of G; the group of all characters of G
$\mathrm{Ann}(\chi)$	Annihilator of character χ: $\{g \in G : \chi(g) = 1\}$, a subgroup of G
$\mathrm{Ann}(H)$	Annihilator of subgroup $H \leq G$: $\{\chi \in \widehat{G} : \chi(h) = 1\ \forall h \in H\}$

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Symbol	Meaning
ev_g	Evaluation character of $\widehat{G}$ defined by $\mathrm{ev}_g(\chi) = \chi(g)$
Maps on Finite Fields	
$\mathrm{Tr}\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute trace function
$N\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute norm function
$\mathrm{Tr}_{E/\mathbb{F}_q}$	Trace function from E to $\mathbb{F}_q$
$\mathrm{N}_{E/\mathbb{F}_q}$	Norm function from E to $\mathbb{F}_q$
$\sigma\colon \mathbb{F}_q \to \mathbb{F}_q$	Frobenius automorphism: $\sigma(\alpha) = \alpha^p$
Additive Characters of $\mathbb{F}_q$	
χ, τ	Generic notation for additive characters
χ_1	Canonical additive character of $\mathbb{F}_q$: $\chi_1(\alpha) = e^{2\pi i\,\mathrm{Tr}(\alpha)/p}$
χ_β	Additive character defined by $\chi_\beta(\alpha) = \chi_1(\beta\alpha)$, for $\beta \in \mathbb{F}_q$
$\boldsymbol{\mathcal{X}}_q$	Set of all additive characters of $\mathbb{F}_q$
$\chi^{(r)}$	Additive character of $\mathbb{F}_q$ lifted to E via $\chi^{(r)} = \chi \circ \mathrm{Tr}_{E/\mathbb{F}_q}$, $[E : \mathbb{F}_q] = r$
Multiplicative Characters of $\mathbb{F}_q$	
ψ, λ	Generic notation for multiplicative characters
ψ_0	Trivial multiplicative character of $\mathbb{F}_q$
η	Quadratic character of $\mathbb{F}_q$ (for odd q): $\eta(\alpha) = \pm 1$ according as α is a square
$\boldsymbol{\mathcal{M}}_q$	Set of all multiplicative characters of $\mathbb{F}_q^*$
$\boldsymbol{\mathcal{M}}_q^{(e)}$	Set of multiplicative characters of $\mathbb{F}_q$ of exponent e
$\mathrm{ord}(\psi)$	Order of ψ: smallest positive integer d with $\psi^d = \psi_0$
$\left(\frac{\alpha}{q}\right)$	Legendre symbol; equals $\eta(\alpha)$ for $\alpha \in \mathbb{F}_q$
$\psi^{(r)}$	Multiplicative character of $\mathbb{F}_q$ lifted to E via $\psi^{(r)} = \psi \circ \mathrm{N}_{E/\mathbb{F}_q}$, $[E\colon \mathbb{F}_q] = r$
Gaussian and Jacobi Sums	
$G(\psi, \chi)$	Gaussian sum: $\sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha)\,\chi(\alpha)$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$
$J_\alpha(\lambda_1, \ldots, \lambda_k)$	Generalized Jacobi sum: $\displaystyle\sum_{\substack{\alpha_1,\ldots,\alpha_k \in \mathbb{F}_q \\ \alpha_1+\cdots+\alpha_k=\alpha}} \lambda_1(\alpha_1)\cdots\lambda_k(\alpha_k)$
$J(\lambda_1, \ldots, \lambda_k)$	Jacobi sum; special case $\alpha = 1$ of J_α
$J_0(\lambda_1, \ldots, \lambda_k)$	Jacobi sum at zero; special case $\alpha = 0$ of J_α
Kloosterman Sums	
$K(\chi; a, b)$	Kloosterman sum: $\sum_{\gamma \in \mathbb{F}_q^*} \chi(a\gamma + b\gamma^{-1})$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $a, b \in \mathbb{F}_q$
$K(\chi^{(r)}; a, b)$	Kloosterman sum lifted to the degree-r extension of $\mathbb{F}_q$
w_1, w_2	Reciprocal roots of the L-function associated to a Kloosterman sum
Salié and Jacobsthal Sums	
$S(\chi, \tau)$	Salié sum: $\sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma)\,\chi(\gamma)\,\tau(\gamma^{-1})$, for $\chi, \tau \in \boldsymbol{\mathcal{X}}_q$, q odd
$H_n(\alpha)$	Jacobsthal sum: $\sum_{\gamma \in \mathbb{F}_q} \eta(\gamma^{n+1} + \alpha\gamma)$, for $\alpha \in \mathbb{F}_q^*$

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Symbol	Meaning
$I_n(\alpha)$	Associated Jacobsthal sum: $\sum_{\gamma\in\mathbb{F}_q} \eta(\gamma^n+\alpha)$, for $\alpha \in \mathbb{F}_q^*$
Classical Riemann Zeta Function	
$\zeta(s)$	Riemann zeta function: $\sum_{n=1}^{\infty} n^{-s}$, for $\Re(s) > 1$
$\xi(s)$	Completed zeta function: $\frac{1}{2}s(s-1)\,\Gamma(s/2)\,\pi^{-s/2}\,\zeta(s)$
$\Gamma(s)$	Gamma function: $\int_0^{\infty} e^{-t}t^{s-1}\,dt$
$\Lambda(n)$	Von Mangoldt function: $\log p$ if $n = p^k$, else 0
ρ	Non-trivial zero of $\xi(s)$
Dirichlet L-functions over $\mathbb{F}_q[X]$	
Φ	Set of all monic polynomials in $\mathbb{F}_q[X]$
Φ_d	Monic polynomials in $\mathbb{F}_q[X]$ of degree exactly d
$\mathrm{Irr}(\Phi)$	Set of irreducible monic polynomials in Φ
$\mathfrak{n}(f) = q^{\deg(f)}$	Norm of the polynomial $f \in \mathbb{F}_q[X]$
G	Group of rational functions $r = h_1/h_2$ in $\mathbb{F}_q(X)$ with $h_1, h_2 \in \Phi$ monic
$\overline{G}$	Subgroup of G on which a given character is defined; extended to G by zero
$L(s,\chi)$	Dirichlet L-function: $\sum_{h\in\Phi} \chi(h)\,\mathfrak{n}(h)^{-s}$, for a multiplicative character χ
$\overline{L}(z,\chi)$	Power series satisfying $\overline{L}(z,\chi) = \sum_{h\in\Phi} \chi(h)\,z^{\deg(h)}$ and $\overline{L}(q^{-s},\chi) = L(s,\chi)$
L_k	k-th log-derivative coefficient: $\displaystyle\sum_{\substack{h\in\mathrm{Irr}(\Phi)\\ \deg(h)\mid k}} \deg(h)\,\chi(h)^{k/\deg(h)}$
Equations over $\mathbb{F}_q$ *— Solution Sets*	
$\mathcal{V}(f)$	Solution set of $f(X_1,\ldots,X_n) = 0$ in $\mathbb{F}_q^n$
$Z(f)$	Number of solutions: $\lvert\mathcal{V}(f)\rvert$
$\overline{\mathcal{V}}(f)$	Set of *non-trivial* solutions (excluding $(0,\ldots,0)$ if it's a zero) of $f = 0$
$\overline{Z}(f)$	Number of non-trivial solutions: $\lvert\overline{\mathcal{V}}(f)\rvert$
$\mathcal{V}(f_1,\ldots,f_k)$	Common solution set: $\mathcal{V}(f_1)\cap\cdots\cap\mathcal{V}(f_k)$
$Z(f_1,\ldots,f_k)$	Count of common solutions: $\lvert\mathcal{V}(f_1,\ldots,f_k)\rvert$
$Z_E(P=0)$	Number of solutions of $P(X_1,\ldots,X_n) = 0$ in E where $P \in \mathbb{F}_q[X_1,\ldots,X_n]$, $[E\colon \mathbb{F}_q] > 1$
Ω_d	Space $\mathbb{F}_q^{\leq d}[X_1,\ldots,X_n]$ of polynomials of total degree at most d
$\omega(d)$	Set of n-tuples $(i_1,\ldots,i_n) \in \mathbb{Z}_0^n$ with $i_1+\cdots+i_n \leq d$
Quadratic Forms	
C_f	Coefficient matrix of a quadratic form f: $(C_f)_{ij} = a_{ij}$ where $f = X^{\top}C_fX$
$\det(f)$	Determinant of the coefficient matrix C_f
$\vartheta : \mathbb{F}_q \to \mathbb{Z}$	Auxiliary function: $\vartheta(0) = q-1$ and $\vartheta(\alpha) = -1$ for $\alpha \neq 0$
Diagonal Equations	
$d_i = \gcd(k_i, q-1)$	Exponent reduced modulo the order of $\mathbb{F}_q^*$, for $a_iX_i^{k_i}$ in a diagonal equation
$M(d_1,\ldots,d_n)$	$\#\left\{(t_1,\ldots,t_n) \in \prod_i[d_i-1] : \frac{t_1}{d_1}+\cdots+\frac{t_n}{d_n} \in \mathbb{Z}\right\}$
$D = \mathrm{lcm}(d_1,\ldots,d_n)$	LCM of exponents; appears in the closed-form formula for $M(d_1,\ldots,d_n)$
Character Sums with Polynomials (Weil Sums)	

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Symbol	**Meaning**
$W(\chi; f)$	Additive Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \chi(f(\alpha))$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi; f)$	Multiplicative Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi, \chi; f, g)$	Mixed Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))\, \chi(g(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$, $f, g \in \mathbb{F}_q[X]$
Lifted Weil Sums (over Extension Fields)	
$E = \mathbb{F}_{q^k}$	Degree-k extension of $\mathbb{F}_q$; $[E : \mathbb{F}_q] = k$
$W^{(k)}(\psi^{(k)}; f)$	Lifted multiplicative Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))$
$W^{(k)}(\chi^{(k)}; g)$	Lifted additive Weil sum: $\sum_{\alpha \in E} \chi^{(k)}(g(\alpha))$
$W^{(k)}(\psi^{(k)}, \chi^{(k)}; f, g)$	Lifted mixed Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))\, \chi^{(k)}(g(\alpha))$
Special L-functions (Weil chapter)	
$\overline{G}$ (Weil context)	The specific subgroup $\{r \in G : h_1(\gamma_i) \cdot h_2(\gamma_i) \neq 0\ \forall i\}$ for roots γ_i of a fixed f
$\varsigma : \overline{G} \to \mathbb{F}_q$	Multiplicative map $\varsigma(r) = \prod_i f(\alpha_i) \cdot \prod_j f(\beta_j)^{-1}$ for $r = \prod(X - \alpha_i)/\prod(X - \beta_j)$
$\varrho : G \to \mathbb{F}_q$	Additive map $\varrho(r) = \sum_i g(\alpha_i) - \sum_j g(\beta_j)$ for a fixed polynomial g
$\xi : G \to \mathbb{C}$	Combined character: $\xi(r) = (\psi \circ \varsigma)(r) \cdot (\chi \circ \varrho)(r)$
$L(s, \xi)$	L-function on ξ: $\sum_{h \in \Phi} \xi(h)\, \mathfrak{n}(h)^{-s}$
$\overline{L}(z, \xi)$	Power series: $\sum_{h \in \Phi} \xi(h)\, z^{\deg(h)}$, related by $\overline{L}(q^{-s}, \xi) = L(s, \xi)$
$L^{(k)}(s, \xi^{(k)})$	L-function on the lifted character $\xi^{(k)}$ over $\mathbb{F}_{q^k}$
$\overline{L}^{(k)}(z, \xi^{(k)})$	Power series form of the lifted L-function
$\Phi^{(k)}$	Set of monic polynomials over $E = \mathbb{F}_{q^k}$
$\mathfrak{n}^{(k)}(h) = q^{k \cdot \deg(h)}$	Norm of $h \in \Phi^{(k)}$ over E; analog of $\mathfrak{n}(h) = q^{\deg(h)}$ for $\mathbb{F}_q$
Absolute Varieties	
$\mathcal{A}$	Variety defined over $\mathbb{F}_q$
$Z_k(\mathcal{A})$	Number of $\mathbb{F}_{q^k}$-rational points of $\mathcal{A}$: $\lvert\{(\alpha_1, \ldots, \alpha_n) \in \mathcal{A} : \alpha_i \in \mathbb{F}_{q^k}\}\rvert$

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Chapter 1

Finite Fields

§ 1.1 Field Extensions

We build up the theory of field extensions following Morandi [Mor96].

The study of field theory is primarily the study of field extensions. The classical problems of ruler-and-compass constructions and the solvability of polynomial equations by radicals were settled by analyzing appropriate field extensions. The central idea is to associate a group—the *Galois group*—to a field extension, converting field-theoretic questions into group-theoretic ones. Since the Galois group of a finite extension is finite, the rich structure theory of finite groups becomes available. In this section we build the foundational vocabulary: degrees, algebraic elements, minimal polynomials, and the key structural results about simple and algebraic extensions.

Degrees and Generated Subfields

Let $F \subseteq K$ are fields. Then K is called a *field extension* of F. We denote it by K/F. Here F is called the *base field*. We can think of K as a vector space over F where the addition of two vectors follows the addition operation of K and the multiplication of any $\alpha \in K$ and $c \in F$ follows the multiplication operation of K, $c \cdot \alpha = c\alpha$. Then we denote the dimension of K over F by

$$[K\colon F] \coloneqq \dim_F(K)$$

This dimension is called the *degree of field extension* of F or simply *degree of* $[K\colon F]$. If $[K\colon F] < \infty$ we call it by a special name, *finite extension*. Finite extensions of finite fields are of very important to us and this report we will work on finite fields and their finite extensions. Let $\alpha \in K$. Then we denote $F(\alpha)$ to be the field generated by F and α following the multiplication and addition operations of K. We will define what it means to be the field generated by F and α in Definition 1.1.1. It is very easy to sea $F(\alpha)$ is also a field, and it contains F but also contained in K.

Let F be any field and suppose X be a formal variable. Then $F[X]$ denotes the polynomial ring over F. We define the rational function field $F(X)$ to be the *fraction ring* of $F[X]$ which consists of all quotients f/g where $f, g \in F[X]$ with $g(X) \neq 0$. Similarly, we can define the polynomial ring $F[X_1, \ldots, X_n]$ over multiple formal variables and their fraction field $F(X_1, \ldots, X_n)$.

Look at the following example. $\mathbb{R} \subseteq \mathbb{C}$ are both fields. But $\mathbb{C}$ is a 2-dimensional vector space over $\mathbb{R}$ as we can take the basis $\{1, i\}$. So $\mathbb{C}/\mathbb{R}$ is a finite extension of degree 2. But $\mathbb{Q}/\mathbb{R}$ is an infinite field extension. On the other hand let $\alpha \in \mathbb{R}$. Then consider the field $\mathbb{Q}(\alpha)$. Now degree of $[\mathbb{Q}(\alpha)\colon \mathbb{Q}]$ depends on what kind of element α is. If $\alpha = \sqrt{2}$ then $[\mathbb{Q}(\sqrt{2})\colon \mathbb{Q}] = 2$ but if $\alpha = \pi$ then $[\mathbb{Q}(\pi)\colon \mathbb{Q}] = \infty$.

Definition 1.1.1: Generated Subfields

Let K/F be a field extension and $S \subseteq K$.

(i) The ring generated by F and S, *written $F[S]$, is the intersection of all subrings of K containing F and S.*

(ii) The field generated by F and S, *written $F(S)$, is the intersection of all subfields of K containing F and S.*

If $S = \{a_1, \dots, a_n\}$ is finite we write $F[a_1, \dots, a_n]$ and $F(a_1, \dots, a_n)$. We call $F(\alpha_1, \dots, \alpha_n)$ finitely generated *field.*

From this definition it easily follows that for any $S \subseteq K$, $F(S)$ is the smallest subfield of K containing F and α.

Let K/F be a field extension. For any $\alpha \in K$ consider the map $\mathrm{ev}_\alpha : F[X] \to K$ defined by $f(X) \mapsto f(\alpha)$. Hence, ev_α is a ring homomorphism.

Observation 1.1. *For any $\alpha \in K$, $\mathrm{Im}(\mathrm{ev}_\alpha)$ is a subring of K and ev_α is F-vector space homomorphism.*

Now we can use this map ev_α to give descriptions of the elements of $F[\alpha]$ and $F(\alpha)$.

Lemma 1.1.1

Let K be a field extension of F and let $a \in K$. Then

$$F[a] = \{f(a) : f(x) \in F[x]\} \quad \text{and} \quad F(a) = \left\{\frac{f(a)}{g(a)} : f, g \in F[x],\ g(a) \neq 0\right\}.$$

Moreover, $F(a)$ is the quotient field of $F[a]$.

Proof: The evaluation map ev_a is a ring homomorphism with image $\{f(a) : f \in F[x]\}$, so this set is a subring of K. Any subring of K containing F and a must contain every $f(a)$, so $F[a] = \{f(a) : f \in F[x]\}$. The quotient field of $F[a]$ is $\{f(a)/g(a) : g(a) \neq 0\}$; it is contained in any subfield of K containing $F[a]$, hence equals $F(a)$. ■

Theorem 1.1.2

Let K be a field extension of F and let $a_1, \dots, a_n \in K$. Then

$$F[a_1, \dots, a_n] = \{f(a_1, \dots, a_n) : f \in F[x_1, \dots, x_n]\},$$
$$F(a_1, \dots, a_n) = \left\{\frac{f(a_1, \dots, a_n)}{g(a_1, \dots, a_n)} : f, g \in F[x_1, \dots, x_n],\ g(a_1, \dots, a_n) \neq 0\right\}.$$

Moreover, $F(a_1, \dots, a_n)$ is the quotient field of $F[a_1, \dots, a_n]$.

Proof: The proof is analogous to that of Lemma 1.1.1, using the multivariate evaluation map $f \mapsto f(a_1, \dots, a_n)$. ■

Theorem 1.1.3

Let K be a field extension of F and let $X \subseteq K$. If $\alpha \in F(X)$, then $\alpha \in F(a_1, \dots, a_n)$ for some $a_1, \dots, a_n \in X$. Therefore,

$$F(X) = \bigcup \{F(a_1, \dots, a_n) : a_1, \dots, a_n \in X\},$$

where the union is over all finite subsets of X.

Proof: Each $F(a_1, \ldots, a_n)$ contains F and lies in $F(X)$, so the union U satisfies $U \subseteq F(X)$. The set U contains F and X. If $\alpha \in F(a_1, \ldots, a_n)$ and $\beta \in F(b_1, \ldots, b_m)$, then $\alpha \pm \beta$, $\alpha\beta$, $\alpha/\beta \in F(a_1, \ldots, a_n, b_1, \ldots, b_m) \subseteq U$, so U is a field. Being the smallest subfield containing F and X, it equals $F(X)$. ■

Algebraic Elements and Minimal Polynomials

Definition 1.1.2: Algebraic and Transcendental Elements

Let K be a field extension of F. An element $\alpha \in K$ is algebraic over F *if there is a nonzero polynomial $f(x) \in F[x]$ with $f(\alpha) = 0$. If α is not algebraic over F, then α is* transcendental over F. *If every element of K is algebraic over F, then K is said to be* algebraic over F, *and K/F is called an* algebraic extension.

Hermite (1873) proved that e is transcendental over $\mathbb{Q}$ and Lindemann (1882) proved the same for π. On the other hand, $\sqrt[n]{r} \in \mathbb{R}$ is algebraic over $\mathbb{Q}$ for any $r \in \mathbb{Q}$ (root of $x^n - r$), as is $i = \sqrt{-1}$ (root of $x^2 + 1$).

Definition 1.1.3: Minimal Polynomial

If α is algebraic over a field F, the minimal polynomial *of α over F is the monic polynomial $p(x)$ of least degree in $F[x]$ for which $p(\alpha) = 0$; it is denoted $\min(F, \alpha)$. Equivalently, $\min(F, \alpha)$ is the monic generator of the kernel of the evaluation homomorphism ev_α.*

The minimal polynomial depends on the base field. For instance, $\min(\mathbb{Q}, i) = x^2 + 1$ while $\min(\mathbb{C}, i) = x - i$, since $i \in \mathbb{C}$ is already in the base field.

Structure of Simple Algebraic Extensions

Theorem 1.1.4

Let K be a field extension of F and let $\alpha \in K$ be algebraic over F. If $p(x) = \min(F, \alpha)$, then:

(i) *$p(x)$ is irreducible over F.*

(ii) *If $g(x) \in F[x]$, then $g(\alpha) = 0$ if and only if $p(x) \mid g(x)$.*

(iii) *If $n = \deg(\min(F, \alpha))$, then the elements $1, \alpha, \ldots, \alpha^{n-1}$ form a basis for $F(\alpha)$ over F, so $[F(\alpha) : F] = \deg(\min(F, \alpha)) < \infty$. Moreover, $F(\alpha) = F[\alpha]$.*

(iv) *$F(\alpha) \cong F[x]/(p(x))$.*

Proof: The first isomorphism theorem gives $F[\alpha] \cong F[x]/(p(x))$. Since $F[\alpha] \subseteq K$ is an integral domain, $(p(x))$ is a prime ideal; in a PID every nonzero prime is maximal, so $F[x]/(p(x))$ is a field and $F[\alpha] = F(\alpha)$. This gives (i) and (iv). Part (ii) follows from $\ker(\mathrm{ev}_\alpha) = (p(x))$.

For (iii): any $b \in F(\alpha) = F[\alpha]$ satisfies $b = g(\alpha)$; dividing gives $g = qp + r$ with $\deg r < n$, so $b = r(\alpha)$ lies in the span of $\{1, \alpha, \ldots, \alpha^{n-1}\}$. A nontrivial F-linear relation among these would give a polynomial of degree $< n$ in $\ker(\mathrm{ev}_\alpha)$, contradicting minimality of p. ■

Example 1.1.1

(i) $\alpha = \sqrt[3]{2}$: the polynomial $x^3 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein at $p = 2$, so $\min(\mathbb{Q}, \sqrt[3]{2}) = x^3 - 2$ and $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$, with $\mathbb{Q}$-basis $\{1, \sqrt[3]{2}, \sqrt[3]{4}\}$.

(ii) $\omega = e^{2\pi i/p}$, p prime: since $\omega^p = 1$ and $\omega \neq 1$, it is a root of the cyclotomic polynomial $\Phi_p(x) = x^{p-1} + \cdots + x + 1$, which is irreducible over $\mathbb{Q}$. Hence $\min(\mathbb{Q}, \omega) = \Phi_p(x)$ and $[\mathbb{Q}(\omega) : \mathbb{Q}] = p - 1$.

Finite and Algebraic Extensions

Lemma 1.1.5

Let $K = F(X)$ be the field of rational functions in X over F. If $u \in K$ with $u \notin F$ let $u = f/g$ where $f, g \in F[X]$ and $\gcd(f, g) = 1$. Let $L = F(u)$. Then

$$[K : L] = \max\{\deg(f), \deg(g)\}$$

Lemma 1.1.6

If $[K : F] < \infty$, then K is algebraic and finitely generated over F.

Proof: Let $\{\alpha_1, \ldots, \alpha_n\}$ be an F-basis of K; then $K = F(\alpha_1, \ldots, \alpha_n)$, so K is finitely generated. For any $\alpha \in K$, the $n+1$ elements $1, \alpha, \ldots, \alpha^n$ span only an n-dimensional space, so they are F-linearly dependent. A nontrivial relation gives a nonzero polynomial of degree $\leq n$ in $F[x]$ vanishing at α, so α is algebraic over F. ■

Theorem 1.1.7 Tower Law

Let $F \subseteq L \subseteq K$ be fields. Then $[K : F] = [K : L] \cdot [L : F]$.

Proof: Let $\{a_i\}_{i \in I}$ be an F-basis of L and $\{b_j\}_{j \in J}$ an L-basis of K. We show $\{a_i b_j\}_{I \times J}$ is an F-basis of K.

Spanning. For $x \in K$, write $x = \sum_j \alpha_j b_j$ with $\alpha_j \in L$, and $\alpha_j = \sum_i \beta_{ij} a_i$ with $\beta_{ij} \in F$; then $x = \sum_{i,j} \beta_{ij} a_i b_j$.

Independence. If $\sum_{i,j} \beta_{ij} a_i b_j = 0$, regroup to get $\sum_j \left(\sum_i \beta_{ij} a_i\right) b_j = 0$. By L-independence of $\{b_j\}$, each $\sum_i \beta_{ij} a_i = 0$, and by F-independence of $\{a_i\}$, each $\beta_{ij} = 0$. ■

Theorem 1.1.8

Let K/F be a field extension. If $\alpha_1, \ldots, \alpha_n \in K$ are algebraic over F, then

$$[F(\alpha_1, \ldots, \alpha_n) : F] \leq \prod_{i=1}^{n} [F(\alpha_i) : F].$$

Proof: By induction on n; the case $n = 1$ is trivial. Set $L = F(\alpha_1, \ldots, \alpha_{n-1})$; by induction $[L : F] \leq \prod_{i=1}^{n-1} [F(\alpha_i) : F] < \infty$. Since α_n is algebraic over F and hence over L, with $\min(L, \alpha_n) \mid \min(F, \alpha_n)$ in $L[x]$, we have $[L(\alpha_n) : L] \leq [F(\alpha_n) : F]$. The tower law gives

$$[F(\alpha_1, \ldots, \alpha_n) : F] = [L(\alpha_n) : L] \cdot [L : F] \leq [F(\alpha_n) : F] \cdot \prod_{i=1}^{n-1} [F(\alpha_i) : F].$$

■

Corollary 1.1.9

Let K/F be a field extension and $\alpha \in K$. Then α is algebraic over F if and only if $[F(\alpha) : F] < \infty$. Moreover, K is algebraic over F if $[K : F] < \infty$.

Proof: If α is algebraic, then $[F(\alpha) : F] = \deg \min(F, \alpha) < \infty$ by Theorem 1.1.4(iii). Conversely, if $[F(\alpha) : F] < \infty$, then α is algebraic by Theorem 1.1.6. The final statement is Theorem 1.1.6. ■

Theorem 1.1.10

Let K be a field extension of F and $X \subseteq K$ such that each element of X is algebraic over F. Then $F(X)/F$ is algebraic. If $|X| < \infty$, then $[F(X) : F] < \infty$.

Proof: For any $\alpha \in F(X)$, Theorem 1.1.3 gives $\alpha \in F(a_1, \ldots, a_m)$ for some $a_1, \ldots, a_m \in X$. By Theorem 1.1.8, $[F(a_1, \ldots, a_m) : F] < \infty$, so α is algebraic over F by Theorem 1.1.6. If X is finite, the same bound gives $[F(X) : F] < \infty$. ■

Theorem 1.1.11 Transitivity of Algebraic Extensions

Let $F \subseteq L \subseteq K$ be fields. If L/F and K/L are algebraic, then K/F is algebraic.

Proof: Let $\alpha \in K$. Since K/L is algebraic, there is a nonzero $p(x) = \sum_{k=0}^{m} a_k x^k \in L[x]$ with $p(\alpha) = 0$. Each a_k is algebraic over F, so by Theorem 1.1.8 the extension $L' = F(a_0, \ldots, a_m)$ satisfies $[L' : F] < \infty$. Since $p(x) \in L'[x]$ and $p(\alpha) = 0$, the element α is algebraic over L' and $[L'(\alpha) : L'] < \infty$. By the tower law,

$$[F(a_0, \ldots, a_m, \alpha) : F] = [L'(\alpha) : L'] \cdot [L' : F] < \infty,$$

so α is algebraic over F by Theorem 1.1.6. ■

Algebraic Closure and Transitivity

Definition 1.1.4: Algebraic Closure in K

Let K be a field extension of F. The set

$$\{\, a \in K : a \text{ is algebraic over } F \,\}$$

is called the algebraic closure of F in K.

Corollary 1.1.12

Let K be a field extension of F and let L be the algebraic closure of F in K. Then L is a field, and therefore is the largest algebraic extension of F contained in K.

Proof: Let $a, b \in L$; both are algebraic over F. By Theorem 1.1.8, $F(a,b)/F$ is finite, hence algebraic by Theorem 1.1.6. Since $a \pm b$, ab, a/b $(b \neq 0)$ all lie in $F(a,b) \subseteq L$, the set L is closed under field operations, so it is a field. It is algebraic over F by definition, and it contains any algebraic extension of F inside K. ■

§ 1.2 Trace and Norm

Chapter 2

Bibliography

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